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Cardiovascular Outcomes with Exenatide in Type 2 Diabetes according to Ejection Fraction: the EXSCEL trial

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Abstract

Aims: GLP-1 receptor agonists (GLP-1 RA) reduce major atherosclerotic cardiovascular events (MACE) and cardiovascular mortality in people with type 2 diabetes (T2D). However, previous studies suggest the effects on heart failure (HF) outcomes vary according to left ventricle ejection fraction (LVEF). We aimed to evaluate the effects of exenatide on cardiovascular events according to LVEF in people with T2D.

Methods and Results: Post-hoc analysis of the Exenatide Study of Cardiovascular Event Lowering (EXSCEL) trial evaluating effects of once-weekly exenatide (EQW) vs. placebo on cardiovascular outcomes according to baseline LVEF (<40% or ≥40%). Outcomes were also evaluated according to NYHA class and obesity. The main outcome was hospitalization for HF (hHF). A treatment-by-LVEF interaction was used. In EXSCEL (N=14752), 4749 participants had LVEF available at baseline; 455 (10%) with LVEF<40%, 4294 (90%) ≥40%. LVEF modified the EQW effect on hHF: HR=1.52 (95%CI=0.95–2.43) in participants with LVEF<40% and HR=0.74 (95%CI=0.55–1.01) in those with LVEF≥40% (p-interaction=0.012). No significant treatment-by-LVEF interactions (p-interaction>0.10) were observed for MACE, cardiovascular death or all-cause mortality. hHF risk was also modified by baseline NYHA class: HR=0.91 (95%CI=0.65–1.27) for NYHA I/II, HR=1.84 (95%CI=0.95–3.59) for NYHA III/IV (p-interaction=0.062), mostly driven by the LVEF<40% subgroup. Obesity did not modify the effects of EQW on hHF.

Conclusions: The EQW effect on hHF was influenced by LVEF, with a potentially decreased risk in participants with LVEF ≥40% and increased risk in those with LVEF <40%. The risk of hHF was particularly high in participants with LVEF

<40% and NYHA III/IV. LVEF did not modify the effect of EQW on atherosclerotic outcomes.

Keywords: GLP-1 receptor agonists; Heart Failure; Ejection Fraction; Type 2 Diabetes.

Introduction

GLP-1 receptor agonists (GLP-1 RA) have been shown to reduce atherosclerotic cardiovascular disease (ASCVD) events and cardiovascular mortality in people with type 2 diabetes (T2D). In a meta-analysis of 8 large randomized controlled trials (RCTs) in people with diabetes, treatment with GLP-1 RA decreased the relative risk of major atherosclerotic cardiovascular events (MACE) by 14% [1]. The SELECT trial showed that GLP-1 RA can also reduce ASVCD events in patients with preexisting cardiovascular disease and overweight or obesity but without diabetes [2].

The effect of GLP-1 RA on heart failure (HF) events is less clear. A meta-analysis of RCTs in diabetes suggests that GLP-1 RA may decrease HF hospitalization in patients without previous HF, but not in patients with previously diagnosed HF [3]. In the STEP-HFpEF DM and the STEP-HFpEF trials, treatment with semaglutide improved the functional capacity and quality of life in patients with HFpEF and obesity, with and without diabetes respectively, along with pronounced weight loss [4,5]. These trials were not powered to show an effect on “hard” outcomes; however, a significantly lower risk of HF hospitalization was observed in a pooled analysis of both trials (hazard ratio [HR] of 0.27, 95% confidence interval [CI], 0.12–0.56) [6]. Similarly, it was recently announced that, in the SUMMIT trial, the dual glucose-dependent insulinotropic polypeptide (GIP) and GLP-1 receptor agonist tirzepatide reduced the risk of HF outcomes in patients with HFpEF and obesity by 38% [7].

The beneficial effects of GLP-1 RA on HFpEF have supported a hypothesis that a similar effect could be observed in patients with HFrEF [8]. However, two small trials in patients with HFrEF, the FIGHT (n=300) and the LIVE (n=241) trials,

showed no protective effects of liraglutide in patients with HFrEF, with signals of increased risk of adverse events. In the FIGHT trial, which included patients with established HF and reduced left ventricle ejection fraction (LVEF) who were recently hospitalized, the risk for hospitalization for HF, emergency department visit, or death was higher with liraglutide compared to placebo (97 [63%] - vs 81 [55%] - events; HR=1.34, 95% CI, 1.00–1.80, p=0.05) [9]. In a post-hoc analysis of the FIGHT, there was an increased risk of hospitalization for HF (124 vs 80 events; p=0.061) and arrhythmic events (39 vs 21 events; p=0.088), particularly among patients with a New York Heart Association (NYHA) class of III/IV.[10] In the LIVE trial, which included patients with clinically stable HFrEF and on optimized HF treatment, there was an increased risk of serious cardiac events with liraglutide compared to placebo (12 [10%] - vs 3 [3%] - events, p=0.04), driven by ventricular tachycardias (4 events, including one death, vs 1 event) and atrial fibrillation requiring intervention (4 vs 2 events) [11]. In the SELECT trial, which included patients in secondary prevention of atherosclerotic events with obesity but without diabetes, semaglutide reduced the risk of HF events by 18% [2]. However, a secondary analysis of the SELECT trial showed that, while the beneficial effects of semaglutide on MACE were independent of LVEF, the effects on HF hospitalization varied according to LVEF: in patients classified as having HFpEF, semaglutide reduced HF hospitalizations, while in those classified as having HFrEF the effect on HF hospitalization had a point estimate hazard ratio above 1.0 [12]. In the FLOW trial, semaglutide reduced HF events (7.5% vs. 9.9% in placebo) in a population with chronic kidney disease (CKD) and T2D [13]. However, no data from the FLOW trial has been published to date regarding the effects stratified by LVEF.

The EXSCEL trial is the only large randomized clinical trial of GLP-1 RA in T2D that collected data on LVEF, providing a unique opportunity to evaluate the effect of GLP-1 RAs on cardiovascular outcomes according to LVEF. A previous prespecified analysis of the EXSCEL evaluated the effects of once-weekly exenatide (EQW) according to HF at baseline [14]. While in those without HF at baseline EQW reduced the combined outcome of hospitalization for HF (hHF) or all-cause death (HR 0.79, 95% CI, 0.68–0.92), in those with HF at baseline there was no observed reduction in hHF or all-cause death (HR 1.05, 95% CI, 0.85–1.29; p-interaction=0.031); a similar pattern was observed for all-cause death. The evaluation of outcomes according to LVEF was not the main focus of that prespecified analysis. Using data from tables published in this sub-analysis, we identified a potentially increased risk of hHF in those with LVEF <40%, which was published in a preliminary report [15]. Given these findings, we used the Vivli Platform for scientific sharing and reuse of clinical research data to request access to the full data of the EXSCEL trial. Here, we publish the full report of the effects of EQW on hHF, atherosclerotic events and mortality according to LVEF in the EXSCEL trial.

Methods

Study population

The EXSCEL trial enrolled 14 752 patients in 35 countries between June 2010 and September 2015. The study design and primary results have been published [14,16]. In brief, EXSCEL randomly assigned patients with T2D (glycated hemoglobin at baseline of 6.5%–10.0%), with or without cardiovascular disease,

to receive EQW at a dose of 2 mg each week, or matching placebo. The study was designed to enroll approximately 70% of participants with prior cardiovascular events including coronary, cerebrovascular, or peripheral vascular events. Key exclusion criteria were a history of type 1 diabetes, 2 or more episodes of severe hypoglycemia during the preceding 12 months, and an estimated glomerular filtration rate <30 mL/min per 1.73 m^2. The primary composite outcome of the trial was the first occurrence of MACE, a composite outcome of cardiovascular death, nonfatal myocardial infarction or nonfatal stroke. The median follow-up time was 3.2 years. In the primary analysis, EQW treatment was associated with a numerically lower risk of MACE compared to placebo, although not reaching statistical significance (HR= 0.91; 95% CI, 0.83–1.00, P=0.06) [16].

The EXSCEL trial protocol was approved by the ethics committee at each participating site and all participants provided written informed consent. Data for this post-hoc analysis was accessed through the Vivli® (Center for Global Clinical Research Data) platform with the Vivli ID: 00009286. The shared dataset excluded subjects who had not given or have withdrawn informed consent, as well as subjects from countries not allowing data reuse based on consent wording. In the intention-to-treat population (all subjects who underwent randomization), the shared dataset did not include 637 of the original 14 752 participants; thus, our analysis was performed on 14 115 participants. Of the 4892 participants reported to have LVEF data available [17], our analysis included 4749 participants.

Evaluation of HF, LVEF and NYHA class

HF status at baseline was reported by the investigators based on clinical data including signs/symptoms of HF, cardiac echocardiography and biomarker data [17]. The most recent assessment of LVEF was reported by study investigators as LVEF categories of LVEF <25%, LVEF 25-39%, LVEF 40-55% and LVEF >55%. The LVEF categories in EXSCEL were defined before the establishment of the current universal definition and classification of HF by LVEF. NYHA functional class [18] was also reported by study investigators in patients classified as having HF at baseline.

Cardiovascular outcomes

The main outcome of this analysis was the time to first event of hHF. As secondary outcomes we evaluated time to MACE; cardiovascular death; all-cause mortality; hHF or cardiovascular death; hHF or MACE; hHF or all-cause mortality; and ventricular tachyarrhythmia. All outcomes were adjudicated by an independent clinical events classification committee.

Adverse events

Prespecified events of clinical interest for which information was collected systematically at every follow-up visit, regardless of seriousness, were pancreatitis, neoplasm, severe hypoglycemia, and expected cardiovascular or diabetes-related complications [14].

Statistical analysis

EXSCEL trial participants with echocardiographic data available were divided into two subgroups according to baseline LVEF (<40% vs. ≥40%). Baseline clinical characteristics were described as mean (standard deviation) or median (percentile 25–75) for continuous variables and number (%) for categorical variables, and compared through Student's t-test, Mann-Whitney and Chi^2 tests, as appropriate. The effect of EQW vs. placebo was tested using Cox models for time-to-first event analyses with treatment group as explanatory variable and LVEF at baseline as stratification variable, with treatment-by-LVEF interaction term in the model. The results are presented separately according to baseline left ventricle ejection fraction (LVEF, <40% vs. ≥40) and baseline NYHA class (NYHA I/II vs III/IV), along with the respective interaction P-value. As an additional analysis, the effect of LVEF according to four LVEF subgroups (<25%, 25-39%, 40-55% and >55%) was also evaluated. The interaction P-value for this analysis considered the four LVEF groups as an ordinal variable. Analyses were performed according to the intention-to-treat principle. The Kaplan-Meier method was used to estimate event rates. The effect modification of NYHA class and obesity on cardiovascular outcomes was also assessed in the subgroups based on baseline LVEF. The effect of EQW on body weight, heart rate, systolic and diastolic blood pressure, HbA1c, and estimated glomerular filtration rate (eGFR) over time, was studied using mixed random effects models with treatment, time, LVEF, baseline value of the variable of interest, treatment-by-time, and treatment-by-LVEF interaction terms in the model; random effects were set at the patient ID level using an unstructured covariance matrix. Adverse events were studied among participants who received at least one dose of the study drug using logistic regression models

with treatment group as explanatory variable and LVEF at baseline as stratification variable, with treatment-by-LVEF at baseline interaction term in the model. An analysis restricted to participants with previous diagnosis of HF at baseline was also performed. Additionally, a competing risks regression, using Fine–Gray method, was used to analyze the risk of hHF accounting for competing risk of all-cause death. A two-sided P-value <0.05 were considered statistically significant for main effects, and a two-sided P-value <0.10 were considered statistically significant for interaction tests due to their reduced statistical power. Analyses were performed using Stata® (StataCorp. 2021. Stata Statistical Software: Release 17. College Station, TX: StataCorp LLC).

Results

Baseline patient characteristics

Of 4749 EXSCEL trial participants with LVEF available, 455 (10%) had a LVEF <40%. Compared with participants with LVEF ≥40%, those with LVEF <40% were more frequently men, had a higher prevalence of previous cardiovascular event, coronary artery disease and of clinical diagnosis of HF (*Table 1*). Additionally, participants with LVEF <40% had lower eGFR, lower systolic blood pressure, lower use of biguanides and calcium channel blockers; and higher use of aspirin, ACE inhibitors, diuretics, beta blockers and MRAs. The baseline characteristics of participants treated with EQW and placebo were well balanced within LVEF groups. Comparison of baseline characteristics of participants with LVEF available vs. LVEF missing is shown in *Supplementary Table 1*. B-type natriuretic peptide (BNP) levels were available for 48 (11%) of patients with LVEF <40% and 214 (5%) of patients with LVEF ≥40% at baseline (values shown in *Supplementary Table 2*).

Effects of EQW vs. placebo on cardiovascular outcomes according to LVEF

The effect of EQW (vs. placebo) on hHF was modified by baseline LVEF: HR 1.52 (95% CI, 0.95–2.43) in participants with LVEF <40% and HR 0.74 (95% CI, 0.55–1.01) in those with LVEF ≥40% at baseline (p-interaction = 0.012). There were no significant interactions (p-interaction >0.10) of LVEF with treatment arm on MACE, cardiovascular death, hHF or cardiovascular death, hHF or MACE, ventricular tachyarrhythmia, or all-cause mortality (*Table 2, Figure 1*). The results were similar when restricting the analysis to participants classified has

having HF at baseline (*Supplementary Table 3*). The analysis of the effect of EQW (vs. placebo) on hHF according to the four LVEF groups supports an effect modification across LVEF categories (P-interaction=0.011) with stepwise risk increment for lower LVEF values i.e., highest risk for participants with LVEF <25% followed by LVEF 25-39%, and a decreased risk for those with LVEF 40-55% and >55% (*Figure 2*). The effect of EQW (vs. placebo) on hHF risk accounting for competing risk of all-cause mortality was similar to the primary analysis *(Supplementary Table 4).*

Effects of EQW vs. placebo on cardiovascular outcomes according to NYHA class and obesity

The risk of hHF was modified by NYHA class at baseline: HR 0.91 (0.65–1.27) in NYHA I/II and HR 1.84 (0.95–3.59) in NYHA III/IV (p-interaction=0.062) (*Table 2*). A similar pattern was also observed for the combined outcome of hHF or MACE. The effect modification of NYHA class on hHF and composite outcomes including hHF was mostly driven by the subgroup with LVEF <40% (*Supplementary Table 5*). While NYHA class did not significantly modify the effect of EQW on any of the evaluated outcomes in those with LVEF ≥40%; in those with LVEF <40%, the risk of hHF or all-cause mortality with EQW (vs. placebo) was increase in NYHA III/IV (HR 1.82, 95% CI, 0.96–3.47) comparing with NYHA I/II (HR 0.78, 95% CI, 0.49–1.23) (p-interaction=0.035). A similar pattern was observed for hHF or CV death and hHF or MACE *(Supplementary Table 5)*. Cardiovascular outcomes were not significantly modified by the presence of obesity at baseline *(Supplementary Table 6).*

Effects on Body Weight, Heart Rate, Blood Pressure, HbA_{1c}, and eGFR according to LVEF

The effects of EQW vs. placebo on body weight (overall least-squares mean difference with EQW vs. placebo: −1.17 kg in those with LVEF <40% and -1.09 kg in those with LVEF ≥40%, p-interaction>0.99) and heart rate (+3.88 bpm in LVEF <40% and +2.30 bpm in those with LVEF ≥40%, p-interaction=0.75) were not modified by baseline LVEF. Similarly, the effects of EQW on HbA_{1c}, systolic and diastolic blood pressure, and eGFR were also not modified by baseline LVEF (*Supplementary Figure 1*).

Adverse events with EQW vs. placebo by baseline LVEF

The incidence of serious adverse events (SAE) was modified by baseline LVEF (P-interaction=0.041): among participants with LVEF <40%, 25.3% treated with EQW had SAE vs. 18.5% with placebo (odds ratio 1.49, 95%CI 0.95–2.34); while in the group of participants with LVEF ≥40%, 18.7% treated with EQW had SAE vs. 20.3% with placebo (odds ratio 0.90, 95%CI 0.78–1.05) (*Table 3*). A similar pattern was observed for premature permanent discontinuation of treatment. *Supplementary Table 7* shows the adverse events of interest, diabetes complications, expected events, and cardiovascular events of interest according to baseline LVEF and treatment arm.

Discussion

In the EXSCEL trial, the effect of EQW on HF outcomes was modified by LVEF, with a potentially protective effect in participants with LVEF ≥40% and increased risk of hHF in participants with LVEF <40%. NYHA class also modified the effect of EQW on hHF, with an increased risk among more symptomatic patients (NYHA class III/IV) particularly driven by those with a LVEF <40%.

Our analysis adds relevant information to the previous literature published from the EXSCEL trial. First, the HF sub-analysis concluded that exenatide reduced HF events and all-cause mortality in those without baseline HF but had a neutral effect for those with HF at baseline [14]. Here, we show that the apparent neutrality of EQW on HF outcomes among those with HF was, in fact, the result of opposing effects: a protective effect in those with preserved LVEF and a harmful effect in those with reduced LVEF. The HF sub-analysis also reported a significant effect modification by HF status on the combined outcome of hHF or all-cause death [14]. Here, we show that the interaction is driven by an increased risk in those with reduced LVEF, particularly those with NYHA class III/IV. These patients experienced a 1.8-fold increased risk (HR 1.82, 95% CI 0.96-3.47) of hHF or all-cause mortality with EQW compared to placebo.

Although we had already reported that LVEF could modify the effects of EQW on hHF [15], the current analysis of the complete data of the EXSCEL trial provides crucial additional information beyond what was identified in the preliminary report. The preliminary report was limited by the lack of data on cardiovascular death, all-cause death, and atherosclerotic events; the use of odds ratios instead of time-to-event analysis; and notably, the absence of information on whether NYHA class could further stratify the risk of adverse

outcomes with GLP-1 RA. In the current analysis, we show that the pattern of effect modification by LVEF is also observed for the combined outcomes of hHF or cardiovascular death and hHF or all-cause mortality, but not for MACE, CV mortality or all-cause mortality. We also show that the risk of hHF events in those with LVEF <40% is particularly increased in NYHA class III/IV, and that this sub-population shows a trend from harm with EQW vs placebo in all studied outcomes, including MACE and all-cause mortality.

The results of this analysis are consistent with previous literature on the effects of GLP-1 RA on HF events [19]. In the FIGHT trial, liraglutide treatment showed a tendency for an increased risk of HF hospitalization and all-cause mortality in patients with advanced HFrEF and recent hospitalization for HF [9,10]. The increased risk with liraglutide was higher among patients in NYHA class III/IV [10]. In both the FIGHT trial and the LIVE trial, treatment with liraglutide was associated with an increased risk of adverse events in patients with HFrEF [9,11], a finding that we also observed with EQW in this analysis of the EXSCEL trial. In contrast, studies evaluating the use of GLP-1 receptor agonists (GLP-1 RA) in patients with HFpEF have suggested a protective effect on HF events [4,5]. A pooled analysis of the STEP-HFpEF and STEP-HFpEF DM trials showed that treatment with semaglutide reduced the risk of HF events, despite the low number of events in both trials [6]. The absence of effect modification on MACE is also consistent with previous literature [3].

In the SELECT trial, semaglutide reduced MACE and cardiovascular mortality independently of LVEF in patients with obesity without diabetes [2]. However, while the risk of HF hospitalization was reduced with semaglutide in patients with HFpEF, in patients classified by the investigators as having HFrEF the

effect on HF hospitalization had a point estimate HR above the unit [12]. Patients classified as having HFrEF in SELECT had an event rate for HF hospitalization or cardiovascular death that was not consistent with HFrEF (approximately 4% per year), suggesting that some patients with investigator-identified HFrEF may not have had HFrEF. Most likely, they had meaningful coronary artery disease with a mild degree of impairment of systolic function. In the SELECT, more than 30% of patients were NYHA class I and only 8.5% were NYHA class III (NYHA IV was not included). Furthermore, given the low event rate of HF events, it is possible that some patients were classified as symptomatic due to exertional symptoms caused by obesity, rather than HF [20]. Still, the direction of the effect of semaglutide was concordant with this and other prior reports using GLP1-RA in HFrEF [20]. Of note, both the SELECT and the EXSCEL trials had atypically low number of HF events in patients with HFrEF comparing to the event rate observed in recent trials that include patients with HFrEF with raised natriuretic peptides (16-21% events per patient/year) [21,22]. As such, both EXSCEL and SELECT trial may not represent the population with HFrEF at higher risk of adverse events with GLP-1 RA [19].

Whether different GLP-1 RA have different effects on HF events is uncertain. EQW is structurally related to exendin-4 and is administered weekly [23]. Others, like liraglutide and semaglutide, are based on human GLP-1 molecule [23]. To date, most meta-analyses have not identified differences on cardiovascular outcomes based on the molecular structure or dosing of administration [1,24]. Exenatide is associated with weight loss that is greater than some GLP-1 RA (e.g. albiglutide) but lower than others (e.g. semaglutide) [23]. The protective effect on atherosclerotic outcomes has been described to be independent of the

effects on weight loss [1,24]. Furthermore, data from trials in T2D do not suggest that the weight loss is determinant for the HF outcomes. In the SUSTAIN-6 trial, which assessed the cardiovascular effects of semaglutide (a GLP-1 RA associated with marked weight loss) in T2D, the HR for hHF was 1.11 (95% CI 0.77-1.61)[25]; while in the HARMONY Outcomes trial, which assessed albiglutide (a GLP-1 RA associated with minimal weight loss), the HR for hHF was 0.71 (95% CI 0.53-0.94)[1]. The differences in HF outcomes between trials in T2D are more likely related to the distribution of LVEF across the population rather than the effects on weight loss of individual drugs.

In this analysis of the EXSCEL trial, approximately one-third of the participants did not have obesity, which differs from recent trials evaluating the effects of GLP-1 RA in patients with HFpEF. The STEP-HFpEF trials and the SUMMIT trial restricted the inclusion of participants with a BMI of at least 30 kg/m^2 [6]. Interestingly, the results of EQW on HF outcomes were not modified by the presence of obesity; this suggests that protective effect of GLP-1 RA in HFpEF may also apply to patients without obesity, and that the harmful effects in patients with HFrEF may be observed in those with and without obesity.

Different mechanisms may explain the observed effects of GLP-1 RA on atherosclerotic events and HF events. GLP-1 RA have been proposed to have direct anti-atherosclerotic effects and indirect effects through improved cardiovascular risk factors [26]. The protective effect on HFpEF events may be explained by a decrease on epicardial mass and inflammation, an improvement of cardiovascular risk factors and/or a reduction of inflammatory markers [27,28]. On the other hand, the adverse effects in patients with HFrEF may be related to the increase of the intracellular levels of cyclic AMP by GLP-1 RA and increased

heart rate [29-31]. While this is not detrimental on patients with HFpEF, increases in intracellular of cyclic AMP levels appear to be central mechanisms of progression of the disease in HFrEF [31]. This is further reinforced by the well-established protective effect of beta-blockers in HFrEF (which reduce intracellular levels of cyclic AMP) and by the detrimental effects of milrinone in HFrEF (which increases intracellular levels of cyclic AMP) [32,33]. The results of this post-hoc analysis also suggest that the population currently classified as having heart failure with mid-range ejection fraction (HFmrEF) may respond more similarly to those with HFpEF than to those with HFrEF when treated with GLP-1 RA. An EXSCEL substudy evaluating proteomic pathways demonstrated that most circulating proteins across various biological domains were similar between HFmrEF and HFpEF, further indicating that HFmrEF may be more biologically similar to HFpEF than to HFrEF [34].

From a clinical perspective, our results suggest that GLP-1 RA can reduce atherosclerotic events and HF hospitalization among those with preserved LVEF (with or without HF) but may increase the risk of HF hospitalization in HFrEF. Whether the balance of reduced risk of atherosclerotic events versus increased risk of HF hospitalization may be beneficial for some patients with HFrEF remains uncertain. This balance may depend on the risk of atherosclerotic events and the severity and control of HFrEF. Trials focusing on the HFrEF population are needed to better understand the effects of the GLP-1 RA class on cardiovascular outcomes in this population.

Our study has limitations that we should acknowledge. This analysis was not pre-specified, and the randomized treatment was not stratified according to baseline LVEF. Furthermore, the primary objective of the EXSCEL trial was to

evaluate the effect of EQW on MACE and several relevant parameters related to HF were not collected. These limitations include a low proportion of patients with natriuretic peptide measurements. LVEF was classified by the investigators in categories, and the exact value was not available; however, LVEF is not a precise measurement and a digit preference for reporting in multiples of 5 has been described [35]. Participants with LVEF <40% represented only a small proportion of the overall population of the EXSCEL trial. In participants with HFrEF, only about one-third were treated with mineralocorticoid receptor antagonists and less than 1% were treated with SGLT-2 inhibitors at baseline. Furthermore, open label drop-in of SGLT-2 inhibitors may also have affected the results [36]. Whether the observed effects with EQW would be the same under optimized guideline-directed medical therapy is unknown. Despite these limitations, this study describes the first analysis of a large RCT in T2D that evaluated the effect a GLP-1 RA according to LVEF. Given the growing role of GLP-1 based therapies in the treatment of diabetes and obesity, understanding how LVEF modifies the effects of GLP-1 RA on cardiovascular outcomes is of major importance. Since this is a secondary analysis, our results do not replace a well-designed RCT specifically aimed at addressing this question. Further large outcome RCTs focused on testing the efficacy and safety of GLP-1 RA in HFrEF patients are needed.

In conclusion, in people with T2D, the effect of EQW on hHF was influenced by LVEF, with a potentially increased risk in participants with LVEF <40% and a decreased risk in those with LVEF ≥40%. The increased risk of hHF with EQW was particularly high in participants with NYHA III/IV, an effect mostly driven by

the subgroup with LVEF<40%. LVEF did not modify the effect of EQW on atherosclerotic events.

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Conflict of interest

JSN reports having received consulting and speaker fees from AstraZeneca, BIAL, Boehringer Ingelheim, Eli Lilly, and Novo Nordisk and speaker fees from Merck. RJM reports having received support and honoraria from AstraZeneca. RRH reports personal fees from Anji Pharmaceuticals, AstraZeneca and Novartis and speaker fees from Lilly. FZ reports personal fees from 89bio, Applied Therapeutics, Bayer, Biopeutics, Boehringer, BMS, CVRx, Cardior, Cambrian, Cereno pharmaceutical, Cellprothera, CEVA, Merck, Novartis, NovoNordisk, Otsuka, Owkin, Servier, having stock options at G3Pharmaceutical and equities at Cereno pharmaceutical, Cardiorenal, Eshmoun Clinical research and being the founder of Cardiovascular Clinical Trialists (CVCT). MP reports personal fees for consulting from 89bio, Abbvie, Actavis, Altimmune, Alnylam, Amarin, Amgen, Ardelyx, AstraZeneca, Attralus, Biopeutics, Boehringer Ingelheim, Caladrius, Casana, CSL Behring,

Cytokinetics, Imara, Lilly, Medtronic, Moderna, Novartis, Pharmacocosmos, Reata, Relypsa, and Salamandra. JB is a consultant to Abbott, American Regent, Amgen, Applied Therapeutic, AskBio, Astellas, AstraZeneca, Bayer, Boehringer Ingelheim, Boston Scientific, Bristol Myers Squibb, Cardiac Dimension, Cardiocell, Cardior, CSL Bearing, CVRx, Cytokinetics, Daxor, Edwards, Element Science, Faraday, Foundry, G3P, Innolife, Impulse Dynamics, Imbria, Inventiva, Ionis, Levator, Lexicon, Lilly, LivaNova, Janssen, Medtronics, Merck, Occlutech, Owkin, Novartis, Novo Nordisk, Pfizer, Pharmacosmos, Pharmain, Prolaio, Pulnovo, Regeneron, Renibus, Roche, Salamandra, Salubris, Sanofi, SC Pharma, Secretome, Sequana, SQ Innovation, Tenex, Tricog, Ultromics, Vifor, and Zoll. JPF is a consultant for Boehringer Ingelheim and AstraZeneca; he has received research support from Novartis, Boehringer Ingelheim, AstraZeneca and Bayer through his institution.

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Figure legends

Figure 1. Kaplan-Meier curves for heart failure hospitalization (A), major atherosclerotic cardiovascular events (B), cardiovascular death (C) and all-cause mortality (D). Hazar ratios and p-interaction are shown in table 2.

Figure 2. Effect of once-weekly exenatide vs. placebo on heart failure hospitalization by baseline left ventricle ejection fraction.

Table 1. Baseline characteristics according to LVEF

Characteristic	LVEF <40% (N=455)		LVEF ≥40% (N=4,294)	
	Placebo (N=206)	Once-weekly exenatide (N=249)	Placebo (N=2,172)	Once-weekly exenatide (N=2,122)
Age, n (%)				
20-50 years	11 (5)	20 (8)	154 (7)	169 (8)
51-60 years	61 (30)	69 (28)	658 (30)	656 (31)
61-70 years	91 (44)	119 (48)	906 (42)	844 (40)
>70 years	43 (21)	39 (16)	454 (21)	454 (21)
Male sex, n (%)	162 (79)	204 (82)	1,486 (68)	1,440 (68)
Smoking, n (%)				
Never	81 (39)	90 (36)	945 (44)	899 (42)
Former smoker	105 (51)	124 (50)	979 (45)	971 (46)
Current smoker	20 (10)	34 (14)	247 (11)	252 (12)
BMI, n (%)				
≤25	29 (14)	20 (8)	132 (6)	132 (6)
25-30	65 (32)	59 (25)	599 (28)	590 (28)
>30	109 (54)	163 (67)	1,424 (66)	1,384 (66)
Systolic BP, mmHg	128 ± 17	129 ± 18	136 ± 17	135 ± 16
Diastolic BP, mmHg	75 ± 11	76 ± 12	77 ± 11	78 ± 10
Heart rate, bpm	72 ± 11	73 ± 10	71 ± 10	72 ± 10
Prior CV event, n (%)	193 (94)	239 (96)	1,905 (88)	1,840 (87)
Coronary artery disease, n (%)	186 (90)	227 (91)	1,639 (75)	1,588 (75)
Cerebrovascular disease, n (%)	32 (16)	39 (16)	380 (18)	349 (16)
Peripheral arterial disease, n (%)	36 (17)	47 (19)	368 (17)	319 (15)
Heart failure, n (%)	128 (62)	169 (68)	561 (26)	499 (24)
NYHA Class*, n (%)				
I	22 (17)	33 (20)	163 (29)	163 (33)
II	74 (58)	93 (55)	331 (59)	271 (54)
III	27 (21)	39 (23)	65 (12)	64 (13)
IV	5 (4)	4 (2)	2 (0.4)	1 (0.2)
Duration of T2D (yrs)	11 (6-18)	11 (7-18)	12 (7-18)	12 (7-18)
Antihyperglycemic agents therapy, n (%)				

Biguanides	126 (61)	151 (61)	1,577 (73)	1,563 (74)
DPP-4 inhibitors	30 (15)	37 (15)	327 (15)	336 (16)
Sulphonylureas	65 (32)	74 (30)	757 (35)	725 (34)
Thiazolidinediones	2 (1)	2 (1)	71 (3)	83 (4)
SGLT-2 inhibitors	1 (1)	0 (0)	6 (0.5)	7 (0.6)
Insulin	102 (50)	137 (55)	1,070 (49)	1,039 (49)
Cardiovascular agents therapy, n (%)				
Aspirin	167 (81)	196 (79)	1,658 (76)	1,636 (77)
ACE inhibitor	126 (61)	161 (65)	1,224 (56)	1,143 (54)
ARB	57 (28)	62 (25)	678 (31)	672 (32)
Diuretic	146 (71)	175 (70)	1,104 (51)	1,072 (51)
Beta blockers	181 (88)	219 (88)	1,583 (73)	1,551 (73)
MRA	73 (35)	85 (34)	166 (8)	166 (8)
CCB	52 (25)	49 (20)	810 (37)	806 (38)
Statins	179 (87)	218 (88)	1,838 (85)	1,811 (85)
HbA_{1c} (%)	8.0 ± 0.9	8.2 ± 1.0	8.1 ± 1.0	8.1 ± 1.0
eGFR (mL/min per 1.73 m^2)	69 ± 24	70 ± 23	75 ± 23	76 ± 23

Results are presented as n. (%), mean ± standard deviation or median (percentile 25th - percentile 75th). * Restricted to patients classified as having heart failure at baseline. Legend: ACE: angiotensin-converting enzyme; ARB: angiotensin II receptor blocker; BMI, body mass index; BP, blood pressure; CCB: calcium channel blockers; CV: cardiovascular; eGFR: estimated glomerular filtration rate; HF: heart failure; LVEF: left ventricle ejection fraction; MRA: mineralocorticoid receptor antagonists; NYHA: New York Heart Association; SGLT-2: Sodium-glucose Cotransporter-2; T2D; Type 2 Diabetes.

Table 2 – Cardiovascular outcomes according to LVEF.

Outcome	Placebo n. (%) IR per 100py	Once-weekly exenatide n. (%) IR per 100py	Hazard Ratio (95%CI)	P-interaction
hHF				0.012
LVEF <40%	27/206 (13) IR=4.7	49/249 (20) IR=7.1	1.52 (0.95-2.43)	
LVEF ≥40%	99/2172 (5) IR=1.4	73/2120 (3) IR=1.1	0.74 (0.55-1.01)	
MACE				0.47
LVEF <40%	72/206 (35) IR=13.2	85/249 (34) IR=12.6	0.96 (0.70-1.31)	
LVEF ≥40%	327/2172 (15) IR=4.9	274/2121 (13) IR=4.1	0.84 (0.71-0.98)	
CV death				0.87
LVEF <40%	41/206 (20) IR=6.5	44/249 (18) IR=5.6	0.84 (0.55-1.29)	
LVEF ≥40%	121/2172 (6) IR=1.7	97/2122 (5) IR=1.4	0.81 (0.62-1.06)	
All-cause mortality				0.27
LVEF <40%	52/206 (25) IR=8.3	62/249 (25) IR=7.9	0.92 (0.64-1.34)	
LVEF ≥40%	194/2172 (9) IR=2.7	140/2122 (7) IR=2.0	0.72 (0.58-0.90)	
hHF or CV death				0.14
LVEF <40%	57/206 (28) IR=9.9	73/249 (29) IR=10.5	1.07 (0.76-1.51)	
LVEF ≥40%	188/2172 (9) IR=2.7	146/2120 (7) IR=2.1	0.78 (0.63-0.97)	
hHF or MACE				0.17
LVEF <40%	79/206 (38) IR=15.0	99/249 (40) IR=15.4	1.03 (0.77-1.39)	
LVEF ≥40%	367/2172 (17) IR=5.5	300/2120 (14) IR=4.5	0.82 (0.70-0.95)	
hHF or all-cause mortality				0.037
LVEF <40%	66/206 (32) IR=11.4	87/249 (35) IR=12.6	1.09 (0.79-1.50)	
LVEF ≥40%	251/2172 (12) IR=3.6	183/2120 (9) IR=2.6	0.73 (0.60-0.89)	
Ventricular tachyarrhythmia				0.63
LVEF <40%	5/206 (2) IR=0.8	13/249 (5) IR=1.7	2.11 (0.75-5.93)	
LVEF ≥40%	8/2172 (0.4) IR=0.1	12/2121 (1) IR=0.2	1.51 (0.62-3.71)	

Legend: CV: cardiovascular; LVEF: left ventricle ejection fraction; IR, incidence-rate per 100 person-years; hHF: hospitalization for heart failure; MACE: major atherosclerotic cardiovascular events.

Table 3 – Cardiovascular outcomes according to NYHA Class

Outcome	Placebo n. (%) IR per 100py	Once-weekly exenatide n. (%) IR per 100py	Hazard Ratio (95%CI)	P-interaction
hHF				0.062
NYHA I/II	73/1055 (7) IR=2.3	63/974 (6) IR=2.1	0.91 (0.65-1.27)	
NYHA III/IV	13/144 (9) IR=3.0	26/164 (16) IR=5.7	1.84 (0.95-3.59)	
MACE				0.30
NYHA I/II	192/1055 (18) IR=6.2	171/974 (18) IR=5.7	0.91 (0.74-1.12)	
NYHA III/IV	43/144 (30) IR=10.5	54/164 (33) IR=12.2	1.16 (0.78-1.73)	
CV death				0.78
NYHA I/II	93/1055 (9) IR=2.8	94/974 (10) IR=2.9	1.04 (0.78-1.39)	
NYHA III/IV	29/144 (20) IR=6.4	31/164 (19) IR=6.2	0.96 (0.58-1.60)	
All-cause mortality				0.87
NYHA I/II	130/1055 (12) IR=3.9	132/974 (14) IR=4.1	1.05 (0.82-1.33)	
NYHA III/IV	36/144 (25) IR=8.0	40/164 (24) IR=8.0	1.00 (0.64-1.58)	
hHF or CV death				0.23
NYHA I/II	145/1055 (14) IR=4.5	136/974 (14) IR=4.4	0.98 (0.77-1.24)	
NYHA III/IV	34/144 (24) IR=8.0	49/164 (30) IR=10.7	1.33 (0.86-2.06)	
hHF or MACE				0.084
NYHA I/II	226/1055 (21) IR=7.4	197/974 (20) IR=6.7	0.90 (0.74-1.09)	
NYHA III/IV	46/144 (32) IR=11.6	64/164 (39) IR=15.1	1.31 (0.89-1.91)	
hHF or All-cause mortality				0.18
NYHA I/II	174/1055 (16) IR=5.4	166/974 (17) IR=5.4	0.99 (0.80-1.22)	
NYHA III/IV	40/144 (28) IR=9.3	58/164 (35) IR=12.6	1.35 (0.90-2.02)	
Ventricular tachyarrhythmia				*
NYHA I/II	9/1055 (1) IR=0.3	9/974 (1) IR=0.3	1.05 (0.42-2.64)	
NYHA III/IV	0/144 (0) IR=0	4/164 (2) IR=0.8	*	

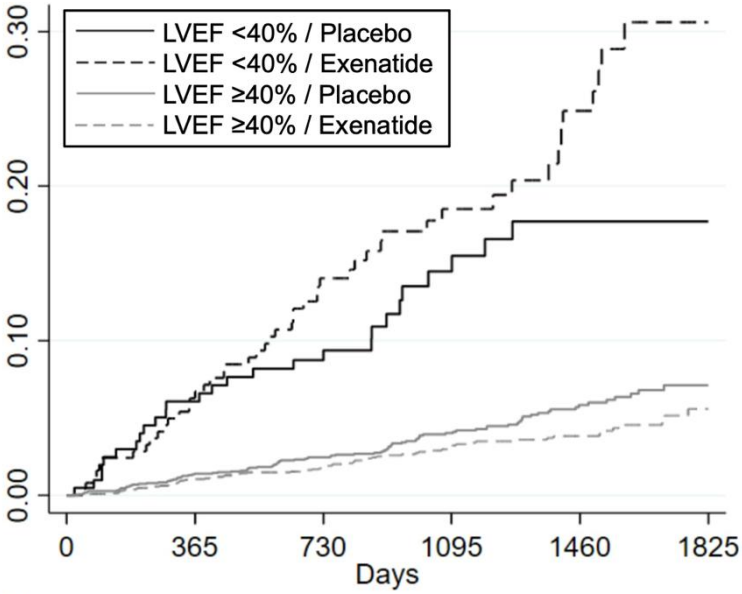
Events are presented as number of events / patients at risk (%).
Legend: CV: cardiovascular; hHF: hospitalization for heart failure; MACE: major atherosclerotic cardiovascular events; NYHA: New York Heart Association.

Table 3 – Severe adverse events, treatment discontinuation and mean percentage of time that participants received the trial regimen according to LVEF

Outcome	Placebo	Once-weekly exenatide	Odds ratio (95%CI)	P-interaction
Any serious adverse event				0.041
LVEF <40%	38/205 (18.5)	63/249 (25.3)	1.49 (0.95-2.34)	
LVEF ≥40%	440/2171 (20.3)	396/2118 (18.7)	0.90 (0.78-1.05)	
Serious adverse event related to trial regimen				*
LVEF <40%	0/205 (0)	4/249 (1.6)	*	
LVEF ≥40%	13/2171 (0.6)	20/2118 (0.9)	1.58 (0.79-3.19)	
Serious adverse event resulting in permanent discontinuation of trial regimen				0.14
LVEF <40%	3/205 (1.5)	8/249 (3.2)	2.24 (0.59-8.54)	
LVEF ≥40%	40/2171 (1.8)	30/2118 (1.4)	0.77 (0.48-1.23)	
Premature, permanent discontinuation of treatment				0.092
LVEF <40%	85/205 (41.5)	117/249 (47.0)	1.25 (0.86-1.82)	
LVEF ≥40%	911/2171 (42.0)	831/2118 (39.2)	0.89 (0.79-1.01)	
Outcome	**Placebo**	**Once-weekly exenatide**	**Difference (95%CI)**	**P-interaction**
Mean percentage of time that participants received the trial regimen				0.17
LVEF <40%	76.3% ± 33.8	71.9% ± 36.1	-4.4% (-10.9 to 2.1)	
LVEF ≥40%	75.6% ± 34.7	76.0% ± 35.9	0.4% (-1.7 to 2.5)	

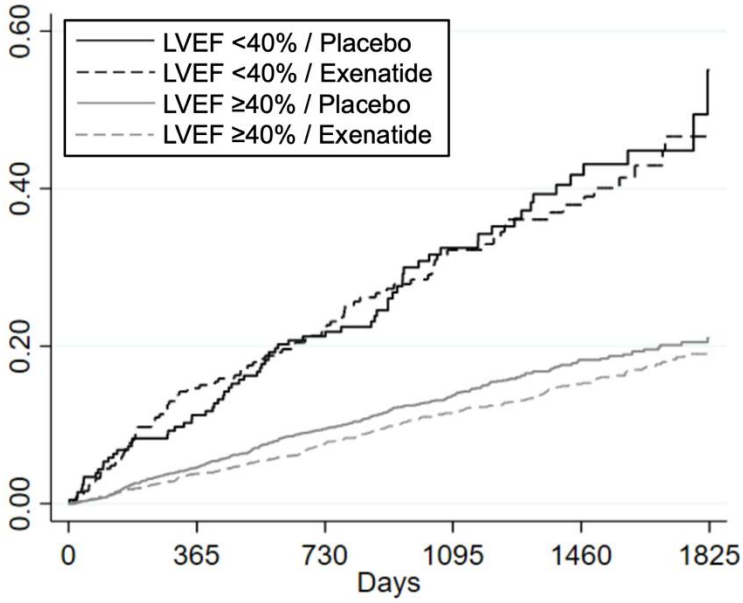
Events are presented as number of events / patients at risk (%).

A. Heart Failure Hospitalization



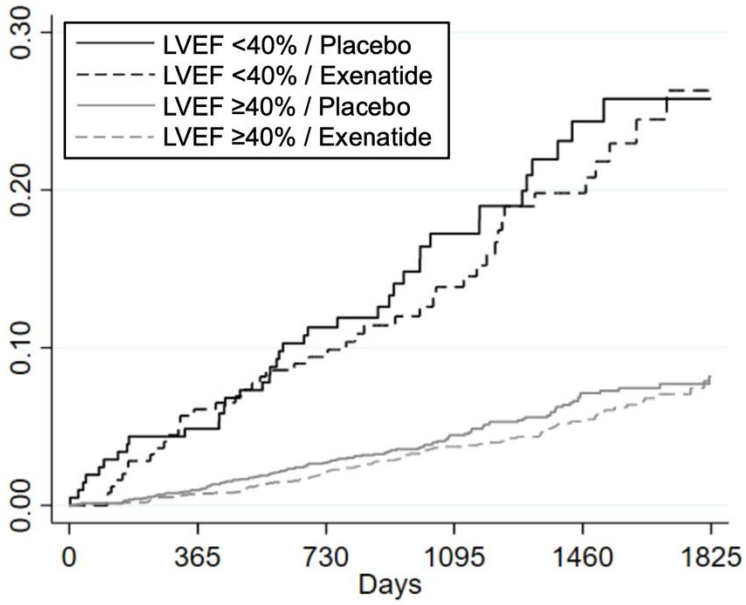
Number at risk						
LVEF <40% / Placebo	206	180	145	85	52	11
LVEF <40% / Exenatide	249	214	170	104	62	15
LVEF ≥40% / Placebo	2172	2089	1834	1157	661	159
LVEF ≥40% / Exenatide	2120	2053	1821	1141	682	151

B. Major Atherosclerotic Cardiovascular Events



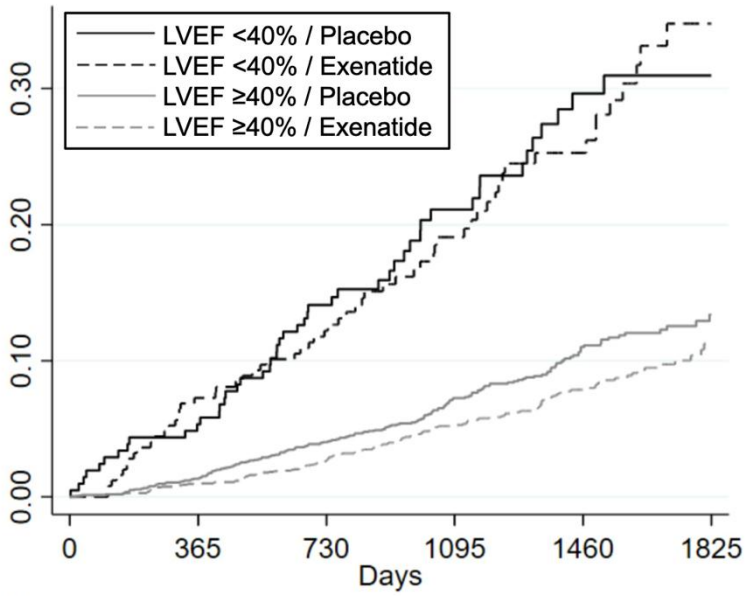
Number at risk						
LVEF <40% / Placebo	206	179	138	78	43	7
LVEF <40% / Exenatide	249	207	170	100	61	14
LVEF ≥40% / Placebo	2172	2040	1746	1075	599	148
LVEF ≥40% / Exenatide	2121	2008	1741	1071	633	142

C. Cardiovascular Death



Number at risk						
LVEF <40% / Placebo	206	195	158	97	58	14
LVEF <40% / Exenatide	249	228	199	128	83	22
LVEF ≥40% / Placebo	2172	2140	1903	1223	713	186
LVEF ≥40% / Exenatide	2122	2098	1881	1203	730	171

D. All-cause Mortality



Number at risk						
LVEF <40% / Placebo	206	195	158	97	58	14
LVEF <40% / Exenatide	249	228	199	128	83	22
LVEF ≥40% / Placebo	2172	2140	1903	1223	713	186
LVEF ≥40% / Exenatide	2122	2098	1881	1203	730	171

Figure 1. Kaplan-Meier curves for heart failure hospitalization (A), major atherosclerotic cardiovascular events (B), cardiovascular death (C) and all-cause mortality (D). Hazard ratios and p-interaction are shown in table 2.

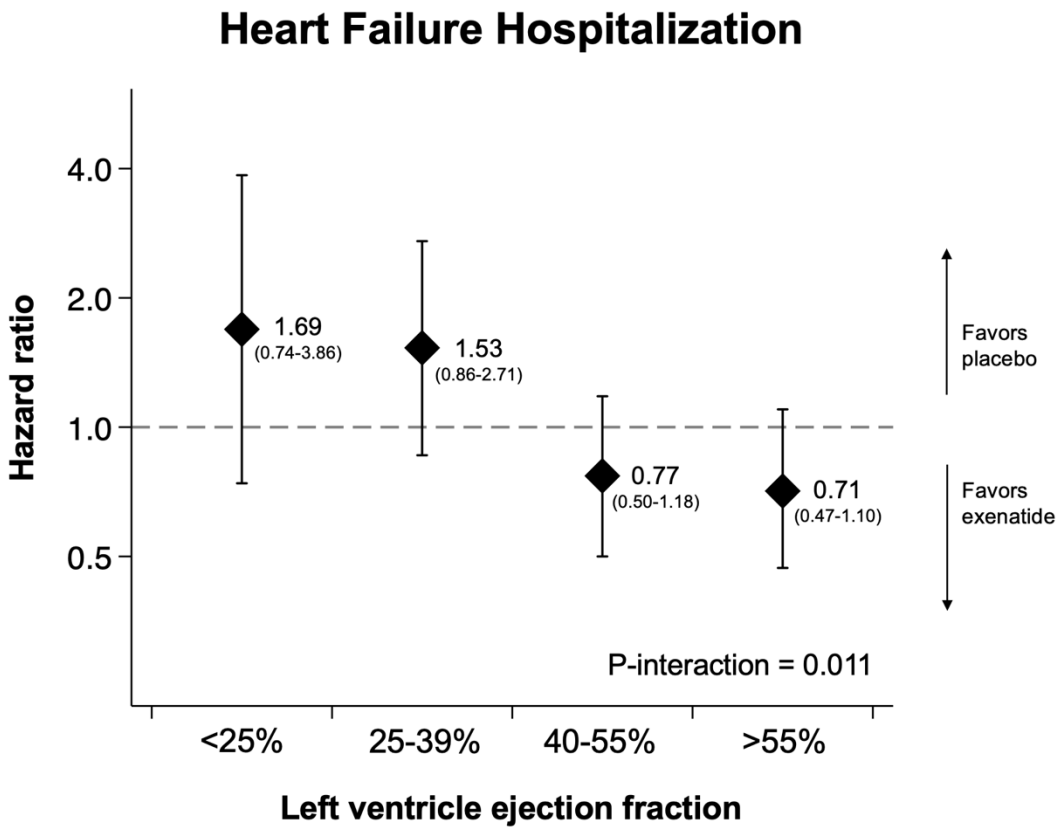


Figure 2. Effect of once-weekly exenatide vs. placebo on heart failure hospitalization by baseline left ventricle ejection fraction.